Quantum Mechanics Assignment-I
CY1001
Jan-May 2026

1. Which of these functions is acceptable as a wave function?
   (a) $\frac{1}{x}(0 \leq x \leq \infty)$
   (b) $\sin(x)\ (-\infty \leq x \leq \infty)$
   (c) $e^{-x}\ (-\infty \leq x \leq \infty)$
   (d) $e^{-x^2}(-\infty \leq x \leq \infty)$

[Answer: d]

2. Which of the operators are nonlinear-
   (a) $\frac{\partial^2}{\partial x^2}$
   (b) $(\ \ )^2$
   (c) $\sqrt{()}$
   (d) $3x^2$

   [Answers: a and d]

3. Consider two operators $\hat{A}$ and $\hat{B}$. Which of the following statements are true regarding their commutativity?

   a. If $\left[\hat{A}, \hat{B}\right] = 0$, then $\hat{A}$ and $\hat{B}$ must share the same eigenstates.
   b. If $\left[\hat{A}, \hat{B}\right] = 0$, then $\hat{A}$ and $\hat{B}$ will have identical eigen values.
   c. If $\left[\hat{A}, \hat{B}\right] = 0$, it implies that the measurement is independent of the order in which the operators act on the function.
   d. If $\left[\hat{A}, \hat{B}\right] = 0$, the measurements of $\hat{A}$ and $\hat{B}$ will not disturb each other

   [Answer: a, c, and d]

4. Determine whether or not the following pairs of operators commute.

$\hat{A}$	$\hat{B}$
(a) $x^2 \frac{d}{dx}$	$\frac{d^2}{dx^2}$
(b) $\frac{d}{dx} - x$	$\frac{d}{dx} + x$
$(c)\ \frac{d^2}{dx^2} - x$	$\frac{d^2}{dx^2} + x$

   [Answer: (a) $-4x\frac{d^2}{dx^2} - 2\frac{d}{dx}$
   (b) 2
   (c) $4\frac{d}{dx}$]

5. Which of these functions are eigenfunctions of the operators $\frac{d}{dx}$ as well as $\frac{d^2}{dx^2}$

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(a) e^{-ax^2} where a is constant
(b) $\cos(bx)$ where b is constant
(c) e^{ikx} where k is constant
(d) $\sin(bx)$ where b is constant

[Answer: (c)]

6. Show that the function f(x) = (8 e^{5x} + 9 e^{-5x}) an eigen function of the operator d^2/dx^2 . What is the eigen value?

   [Answer: 25]

7. For a quantum mechanical system, $\Psi = cos\left(\frac{\pi x}{L}\right)$ for $-\frac{L}{2} < x < \frac{L}{2}$. Normalize Ψ.

   [Answer: $\sqrt{\frac{2}{L}}$]

8. Consider a particle in a quantum state Ø that is the superposition of two eigen functions of energy $\emptyset_1$ and $\emptyset_2$ with energy eigen values E_1 and E_2:

   $\emptyset = C_1\emptyset_1 + C_2\emptyset_2$

The eigen functions, $\emptyset_1$ and $\emptyset_2$, are orthonormal to each other.

What is the probability of measuring E_1 or E_2? What is the average energy?

[Answer: Probability of measuring E1 = $\frac{|C_1|^2}{|C_1|^2+|C_2|^2}$

Probability of measuring E2 = $\frac{|C_2|^2}{|C_1|^2+|C_2|^2}$

Average energy $\langle E\rangle = \frac{|C_1|^2E_1+|C_2|^2E_2}{|C_1|^2+|C_2|^2}$]

9. The speed of an electron is 995 km/s. If the uncertainty in its momentum is to be reduced to 0.01 percent, what uncertainty in its position must be tolerated?

   [Answer: 5.829×10^{-5} cm]

10. A particle is confined to a 3D box with sides of length $L_x = L$, $L_y = 2L$, and $L_z = 4L$. Calculate energy of the first five successive energy levels. Which state(s) is/are degenerate?

    [Answer: $E_1 = (1,1,1), E_2 = (1,1,2), E_3 = (1,2,1), E_4 = (1,1,3), E_5 = (2,1,1)\ and\ (1,2,2)$]

11. Calculate the probability of finding a particle of mass *m* in a 1D box of length L between the length L/4 and L/2 in state n = 1. [Answer: 0.409]

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12. For a particle in 1-D box of length L, show that the Heisenberg's uncertainty principle with respect to position and momentum holds true.

For a particle confined in a one-dimensional box of length L ($0 \leq x \leq L$), the normalized wavefunction is

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right),$$

where $n = 1, 2, 3, \ldots$.

To verify Heisenberg's uncertainty principle, we calculate the uncertainties in position and momentum.

1. Uncertainty in Position

The expectation value of position is

$$\langle x \rangle = \int_0^L x|\psi_n(x)|^2 dx$$

which gives

$$\langle x \rangle = \frac{L}{2}.$$

The expectation value of x^2 is

$$\langle x^2 \rangle = \int_0^L x^2|\psi_n(x)|^2 dx$$

and evaluates to

$$\langle x^2 \rangle = L^2\left(\frac{1}{3} - \frac{1}{2n^2\pi^2}\right).$$

Therefore,

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2$$

$$= L^2\left(\frac{1}{3} - \frac{1}{2n^2\pi^2} - \frac{1}{4}\right)$$

$$= L^2\left(\frac{1}{12} - \frac{1}{2n^2\pi^2}\right).$$

Hence,

$$\boxed{\Delta x = L\sqrt{\frac{1}{12} - \frac{1}{2n^2\pi^2}}}.$$

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2. Uncertainty in Momentum

The momentum operator is

$$\hat{p} = -i\hbar\frac{d}{dx}.$$

The expectation value of momentum is

$$\langle p \rangle = \int_0^L \psi_n^* \left(-i\hbar\frac{d}{dx}\right)\psi_n \, dx = 0.$$

The expectation value of p^2 is

$$\langle p^2 \rangle = \int_0^L \psi_n^* \left(-\hbar^2\frac{d^2}{dx^2}\right)\psi_n \, dx.$$

Since

$$\frac{d^2\psi_n}{dx^2} = -\left(\frac{n\pi}{L}\right)^2 \psi_n,$$

we obtain

$$\langle p^2 \rangle = \frac{n^2\pi^2\hbar^2}{L^2}.$$

Therefore,

$$(\Delta p)^2 = \langle p^2 \rangle - \langle p \rangle^2 = \frac{n^2\pi^2\hbar^2}{L^2},$$

and

$$\boxed{\Delta p = \frac{n\pi\hbar}{L}}.$$

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3. Product of Uncertainties

Multiplying the two uncertainties,

$$\Delta x\,\Delta p = n\pi\hbar\sqrt{\frac{1}{12}-\frac{1}{2n^2\pi^2}}.$$

This can be written as

$$\boxed{\Delta x\,\Delta p = \hbar\sqrt{\frac{n^2\pi^2}{12}-\frac{1}{2}}}.$$

For the ground state ($n=1$),

$$\Delta x\,\Delta p = \hbar\sqrt{\frac{\pi^2}{12}-\frac{1}{2}} \approx 0.568\,\hbar.$$

Since

$$0.568\,\hbar > \frac{\hbar}{2} = 0.500\,\hbar,$$

we have

$$\boxed{\Delta x\,\Delta p > \frac{\hbar}{2}}.$$

For higher values of n, the product becomes even larger.